Intelligent YOLOv8-Based Rover for Precision Agriculture: Tomato Ripeness and Disease Detection

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Abstract*—Tomato cultivation has traditionally relied on manual inspection and generalized irrigation practices, often resulting in inefficient resource use and reduced yield quality. This work presents a compact rover designed to support precision agriculture in tomato farming. The rover is equipped with an OV2640 camera module, a humidity sensor, and a water-level sensor. The OV2640 captures images of tomato fruits and leaves, transmitting them via Wi-Fi to a laptop for analysis. Two custom-trained YOLOv8 deep learning models are employed for visual diagnostics: one determines tomato ripeness, enabling optimal harvest timing, and the other detects common leaf diseases, including Early Blight, Late Blight, Leaf Miner, Leaf Mold, Mosaic Virus, Septoria, Spider Mites, and Yellow Leaf Curl Virus. In addition to visual inspection, the rover measures environmental parameters such as soil moisture and ambient humidity, supporting data-driven irrigation decisions and early preventive measures. Communication between the rover and the processing unit is achieved through live video streaming from the ESP32-CAM, with processed results enabling either manual teleoperation or potential future autonomous navigation. By integrating AI-based plant health assessment with environmental monitoring, the proposed system offers a cost-effective and scalable solution to improve crop quality, optimize resource usage, and enhance decision-making in tomato farming operations.

Keywords*—Precision agriculture, Tomato disease detection, YOLOv8, ESP32-CAM, IoT-based monitoring, Ripeness classification.

I. INTRODUCTION

Tomatoes are one of the most widely cultivated crops across the globe due to their versatility, nutritional value, and economic significance. Their production has steadily increased to meet the rising global demand, supported by advancements in agricultural practices and automation technologies [3], [4]. However, tomato cultivation is still vulnerable to various environmental factors, including soil moisture imbalance, humidity variations, and plant diseases, which can significantly reduce yield quality and quantity [5], [6]. Timely detection of crop health issues, along with effective resource management, plays a crucial role in improving farm productivity and sustainability.

In recent years, the integration of IoT devices and machine learning has transformed the agricultural sector by enabling precise monitoring and decision-making [7], [8]. Devices such as the ESP32-CAM with the OV2640 sensor have become popular due to their low cost, compact design, and wireless connectivity, making them ideal for field deployment in resource-constrained environments [8], [9]. By capturing high-resolution images of crops and transmitting them in real-time, these devices support advanced image-based analysis for agricultural applications.

One of the most pressing challenges in tomato farming is early detection of plant diseases, as many pathogens can spread rapidly if left untreated. Deep learning models, particularly those based on the YOLO architecture, have shown remarkable performance in object detection tasks, including crop disease classification [10]–[12]. Studies have explored disease identification using methods ranging from traditional image processing [13]–[15] to advanced deep neural networks [16]–[20]. Diseases such as Early Blight, Late Blight, Leaf Miner, Leaf Mold, Mosaic Virus, Septoria, Spider Mites, and Yellow Leaf Curl Virus are among the most common threats to tomato crops [16], [19].

In addition to disease detection, monitoring environmental conditions such as soil moisture and humidity is essential for optimizing irrigation schedules and preventing water stress [6], [7]. Smart agriculture systems have been developed to automate irrigation control based on real-time sensor data, reducing water wastage and improving crop growth [6], [7]. Combining environmental monitoring with AI-powered disease detection provides a comprehensive approach to farm management, enabling proactive intervention before problems escalate.

The proposed work presents a small autonomous rover equipped with an OV2640-based ESP32-CAM, humidity and water-level sensors, and dual YOLOv8 models trained from scratch. One model determines tomato ripeness, while the other identifies the presence of specific diseases from leaf images. Data collected from the rover allows

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farmers to monitor crop health and adjust irrigation strategies. The ESP32-CAM streams live video over Wi-Fi to a Python-based processing unit, which analyzes frames using OpenCV and the trained YOLOv8 models. Based on the analysis, control commands are sent back to the rover through Wi-Fi or serial communication, enabling both manual teleoperation and the potential for future autonomous control without major hardware changes.

II. LITERATURE REVIEW

The advancements in modern technology and medicine have profoundly impacted human longevity. Globally, life expectancy saw a significant rise between 1800 and 2001 [1], further extending to 71.4 years by 2015 [2]. This extended lifespan coincides with a dramatic surge in global population, escalating from 1.65 billion in 1900 to approximately 7.9 billion currently [3]. Projections indicate this number will reach an estimated 9.7 billion by 2050 [4]. This accelerating demographic expansion underscores the critical challenge of ensuring sustainable long-term food production. We must cultivate more intelligently. Achieving this necessitates the optimization of every facet of food production. Considerations must extend to land utilization, water consumption, and fertilizer application, as these vital resources are becoming increasingly constrained with the expanding population. Notably, agriculture alone accounts for an estimated 70% of the world's freshwater supply, profoundly affecting the global water system [5]. To address these challenges and enable more precise resource management, innovative solutions like robotic systems for agricultural monitoring are being developed. For instance, research has investigated robotic systems designed to accurately insert soil moisture probes to collect temperature and moisture data across agricultural fields. A key component in such systems is often a highly dexterous 6-degrees-of-freedom robotic manipulator attached to a wheeled base, allowing for precise positioning and data acquisition [6]. Beyond data collection, efficiency in resource use is also a focus; this includes fundamental information and helpful suggestions for using soil moisture sensors to plan irrigation effectively. In a related context of energy management, studies also highlight that significant electricity loss in typical lighting systems can be prevented with automatic control employing Light Dependent Resistors (LDRs)[7]. Further insights into camera module performance, crucial for vision-based applications, come from studies on the ESP32-CAM module. Research has demonstrated its capabilities by collecting 15-minute video streaming logs for various resolutions, with video streamed over WiFi via an HTTP server. A client can access this server on the local WiFi network using the module's IP address. Real-time statistics, including instantaneous FPS, average FPS, runtime, bytes transmitted, and chip temperature, were collected via the UART serial interface and processed using tools like Node-RED. The firmware, based on Arduino's Camera Server example, was modified to stream additional parameters and dynamically control image resolution over UART or HTTP. Performance analysis revealed that resolutions below 320x240 consistently achieve over 50 FPS, but performance degrades significantly at higher resolutions, dropping below 10 FPS at 640x480 and nearing 1 FPS at 1600x1200. This highlights an increasing FPS variability and performance instability with higher computational loads. Interestingly, the operating temperature remained relatively stable (ranging from 69.73°C to 74.18°C) across all resolutions, showing no direct correlation with increasing frame size. The study also noted that the maximum transmission volume was achieved at 400x296 resolution, emphasizing the trade-off between frame size and real-time performance in resource-constrained environments. Given these computational considerations, particularly for embedded systems, our proposed approach for plant disease detection will leverage the capture and processing of discrete still images rather than continuous video streams, aiming to optimize resource utilization while ensuring sufficient visual data for accurate diagnosis [8]. Specifically, when considering the OV2640 camera type commonly used with the ESP32-CAM, research indicates that not all camera types or resolutions are equally suitable for every image recognition application. For instance, resolutions like CIF and QVGA are noted for their faster average speeds (40 ms and 19 ms respectively), making them more compatible for rapid image recognition and face detection tasks. The successful application of the ESP32-CAM in face recognition, where images are fed to a YOLO model in Python to control a relay based on detection results, demonstrates its capability for real-time image processing on an embedded platform. This inherent suitability for tasks like face recognition, which require precise visual data capture and efficient processing, suggests that these types of cameras are also well-suited for agricultural applications such as detecting leaf diseases and assessing the ripeness or overall condition of tomatoes [9]. Beyond these specific camera applications, various machine learning techniques have been explored for automated plant disease detection. One such approach involved a deep learning-based method that utilized a collected dataset of healthy and diseased leaf images to train a convolutional neural network (CNN) model. After training and selecting the best-performing architecture based on validation dataset accuracy, the final model achieved a high accuracy of 99.2% on a separate test dataset. This demonstrates a fast and efficient method for identifying and diagnosing plant leaf diseases, offering a valuable tool for plant pathologists and farmers to quickly and accurately assess plant health [10]. Another study also utilized deep learning, specifically CNNs, to achieve an automatic plant disease detection system based on a simple classification mechanism that exploits CNN's feature extraction capabilities. This research, conducted using a publicly accessible collection of 70,295 images alongside 100 images from experimental and actual environments, achieved an overall 98% testing accuracy on the public dataset and performed well on images of plants from Sukkur IBA University. The high accuracy concluded that CNNs are highly suitable for automatic detection and diagnosis of plants, suggesting integration into mini-drones for live disease detection in cultivated areas. While trained on the Plant Village dataset with 38 classes, the system could determine if a plant had a disease, as

symptoms often share similarities across plant types. Future work proposed a three-layer approach: first detecting the presence of a plant, then identifying the plant type, and finally determining the disease type [11]. In a comprehensive approach to plant leaf disease detection, a method employing efficient image processing and traditional machine learning algorithms was proposed. This methodology involved initial image pre-processing steps such as resizing, histogram equalization for contrast enhancement, color denoising using Gaussian filters in CIELAB color space, and threshold-based segmentation to effectively remove image backgrounds. Following this, salient features including Hu moments (for shape description), Haralick textures, and color histograms (derived from the HSV color space) were meticulously extracted from the pre-processed images. These comprehensive feature sets were then utilized to train and validate various traditional machine learning classifiers, specifically Logistic Regression, Linear Discriminant Analysis, K-nearest neighbor, Decision Tree, Random Forest, and Support Vector Machine algorithms. The study rigorously employed K-fold cross-validation and hyperparameter optimization via GridSearchCV to ensure optimal performance, ultimately demonstrating the efficacy of combining advanced image processing with traditional machine learning techniques for accurate plant disease identification [12]. Another CNN-based model were developed for plant leaf disease detection using image data from crops such as Apple, Corn, Potato, and Tomato sourced from the PlantVillage dataset. The proposed architecture incorporated four convolutional layers, four max-pooling layers, and a fully connected layer. Recognizing the significant impact of disease on leaf texture, the authors integrated Local Binary Patterns (LBP) for texture feature extraction. These texture features were stacked with the original RGB image, forming a four-channel input to the CNN model. Additionally, GrabCut segmentation and Otsu's thresholding were used to effectively remove background details, enhancing the focus on disease-related patterns. The model, trained for 300 epochs, demonstrated high classification accuracy ranging from 95.84% to 98.73%, with texture information contributing up to a 5.11% improvement in accuracy. This approach outperformed several pre-trained CNN architectures and existing texture-based methods. The findings confirm that augmenting CNNs with texture and segmentation techniques significantly boosts the model's disease recognition capability [13]. Another method employed for plant disease detection involves the use of statistical image processing combined with classical machine learning algorithms. In this approach, Random Forest (RF) classifiers were utilized to overcome the overfitting challenges associated with individual decision trees. RF, being part of ensemble learning, leverages multiple decision trees trained on various subsets of the dataset, thereby improving generalization and overall accuracy. The dataset was split into a training set (80%) and a testing set (20%), and a k-fold cross-validation strategy was implemented to obtain unbiased accuracy estimates. The model's performance was evaluated using metrics such as accuracy, F1-score, precision, and recall, with the proposed system achieving an average accuracy of 93% and an F1 score of 0.93. Additionally, ROC curves and confusion matrices were used to visualize false positive and false negative rates. This method demonstrates that machine learning models integrated with traditional image preprocessing techniques offer a computationally efficient alternative to deep learning models, while still maintaining competitive accuracy for plant disease detection tasks [14]. Another method explored for plant disease identification utilized a combination of statistical colour and texture features extracted from segmented leaf images. Specifically, the approach employed Gray Level Co-occurrence Matrix (GLCM) to derive 22 texture features, alongside six colour features, which together formed the input for classification using a Support Vector Machine (SVM). This method achieved an accuracy of 98.79% using 10-fold cross-validation and demonstrated strong performance even on real-world data. The approach illustrates that traditional feature extraction and classical classifiers can be effective without relying on deep learning models like YOLO [15]. In another study focusing on object detection in agriculture, the YOLOv4 model was utilized in conjunction with transfer learning for identifying plant diseases across multiple species. The researchers leveraged a modified YOLOv4 architecture pre-trained on COCO, fine-tuned using annotated leaf datasets to improve domain-specific performance. Data augmentation techniques were applied to improve generalization and robustness. The model demonstrated a high mean average precision (mAP) of 96.46% and was capable of real-time inference, making it highly suitable for deployment in low-power field monitoring devices. The study concluded that YOLOv4 offers a strong balance of accuracy and speed, especially when combined with transfer learning strategies tailored for plant phenotyping and disease detection [16]. An additional study explored the deployment of the YOLOv5 algorithm for efficient and accurate plant disease detection. The model was trained using an annotated dataset of diseased and healthy leaf images, where real-time object detection capabilities were leveraged to localize and classify leaf diseases simultaneously. YOLOv5, implemented in PyTorch, showed notable improvements in speed and precision compared to traditional models, making it ideal for edge-device deployment in agricultural monitoring systems. The study confirms YOLOv5's potential in automating crop health assessments in smart farming applications [17]. Another deep learning-based method for real-time plant disease detection utilized the YOLOv5 architecture, focusing on tomato crops with diseases like early blight and late blight. The dataset comprised over 6,000 labeled images, preprocessed through annotation in Roboflow and augmented using techniques such as flipping and brightness adjustment to improve generalization. The YOLOv5 model was trained with an input resolution of 640×640, using stochastic gradient descent (SGD) and a cosine learning rate scheduler. The system achieved a high mAP@0.5 of 98.3%, precision of 97.8%, and recall of 97.5%, demonstrating strong performance for accurate and fast disease identification. The paper concluded that YOLOv5's lightweight architecture and high-speed inference make it well-suited for real-time agricultural

monitoring applications, especially on edge devices like the ESP32-CAM or Jetson Nano [18]. Another refined approach extended the YOLOv5 architecture by incorporating a hybrid attention module into the feature prediction layers and improving the loss function to enhance disease detection performance, specifically targeting tomato brown rot in greenhouse environments. Although this enhancement introduced a slight decrease in detection speed, the improved model achieved higher average accuracy than both the standard YOLOv5 and Faster R-CNN architectures. The system enabled real-time, highly accurate disease localization, proving especially beneficial for growers with limited expertise. The authors also suggested future directions such as optimizing CNN structures, implementing early disease detection models in complex environments, and exploring federated learning for scalable agricultural applications [19]. A recent study introduced an optimized lightweight YOLOv8n–GGi architecture for apple leaf disease detection by integrating GhostConv and C3Ghost modules to replace conventional convolutional structures. Additionally, a Global Attention Mechanism (GAM) and an improved BiFPN were employed to enhance the model's feature extraction and fusion capabilities. The improved model achieved a precision of 86.3%, recall of 80.2%, and mAP@50 of 86.9%, marking improvements of 3.3%, 2.3%, and 3.4%, respectively, over the baseline YOLOv8. Furthermore, the final model size was reduced to 3.8 MB, representing a 39.7% weight reduction, making it highly suitable for deployment on resource-constrained agricultural devices. These modifications resulted in both improved detection accuracy and computational efficiency, demonstrating the viability of YOLOv8–GGi for real-world smart agriculture scenarios [20].

III. METHODOLGY

A. System Architecture

The proposed system integrates Internet of Things (IoT) sensing, computer vision, and wireless communication into a rover-based platform to support precision agriculture in tomato cultivation. The rover is designed to operate in both greenhouse and open-field environments, acquiring visual and environmental data during field traversal. An OV2640-based ESP32-CAM module captures high-resolution images of tomato fruits and leaves and transmits them via Wi-Fi to a laptop for real-time analysis. Two custom-trained YOLOv8 deep learning models are executed on the laptop, where one model classifies the ripeness stage of tomatoes and the other identifies specific tomato leaf diseases. At the same time, the onboard humidity and water-level sensors collect environmental data to support data-driven irrigation decisions. All subsystems are interconnected through a microcontroller-based control unit that manages sensor readings, image capture, and communication. The overall workflow of the system is illustrated in the system architecture flowchart shown in **Fig. 1.**

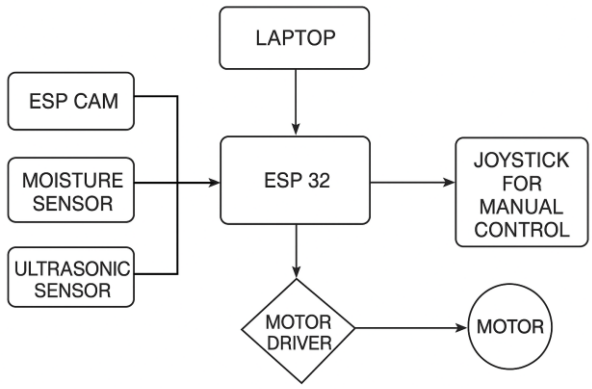

Fig.1 – Block Diagram of the System

B. Hardware Implementation

The hardware framework of the proposed rover consists of sensing, processing, and actuation units integrated into a compact chassis. The primary vision sensor is the OV2640-based ESP32-CAM module, which provides both image acquisition and wireless transmission capabilities. Environmental monitoring is achieved through a capacitive soil moisture sensor for water content measurement and a digital humidity sensor for ambient humidity detection. These sensors are interfaced with a microcontroller that serves as the central control unit for data acquisition and communication.

The rover chassis is constructed to accommodate the sensors, electronics, and power supply while maintaining stability during field operation. A rechargeable battery powers the system, and onboard voltage regulation ensures stable operation of all components. Data from the ESP32-CAM is transmitted over Wi-Fi to a laptop, where the YOLOv8 models perform tomato ripeness classification and disease detection. The control unit is designed to support bidirectional communication, enabling future integration of autonomous navigation modules

C. Data Acquisition and Preprocessing

The rover collects two main types of data visual imagery and environmental measurements. The OV2640 camera module on the ESP32-CAM captures high-resolution images of tomato fruits and leaves under natural lighting conditions. These images, along with readings from the soil moisture sensor and digital humidity sensor, are transmitted via Wi-Fi from the rover to the laptop.

Once the data is received, it is processed by two custom-trained YOLOv8 models. The first model determines the ripeness stage of tomatoes, while the second identifies plant diseases, including Early Blight, Late Blight, Leaf Miner, Leaf Mold, Mosaic Virus, Septoria, Spider Mites, and Yellow Leaf Curl Virus. The models operate in real time, analyzing incoming frames and sensor values to generate actionable results.

The output from the models is then sent back to the rover over the same Wi-Fi connection. This feedback can be used for immediate decisions, such as triggering irrigation or navigation responses, and also logged for long-term crop monitoring. This closed-loop communication

between the rover and the laptop enables a responsive and scalable precision farming system.

D. AI Model Development

The rover's AI component employs a YOLOv8 deep learning architecture to classify tomato ripeness stages and detect major leaf diseases. The dataset was prepared by collecting and annotating images of tomato fruits and leaves under various lighting and environmental conditions. Nine categories were considered during training: Early Blight, Late Blight, Leaf Miner, Leaf Mold, Mosaic Virus, Septoria, Spider Mites, Yellow Leaf Curl Virus, and Healthy Before training, the dataset underwent preprocessing, including image resizing, normalization, and augmentation (rotation, flipping, and brightness variation) to enhance model robustness against environmental variations. The YOLOv8 model was selected for its capability to balance detection accuracy with real-time processing requirements on embedded systems. The model was trained from scratch, and the loss curves for training and validation, along with precision, recall, and mAP metrics, are presented in **Fig. 2**.

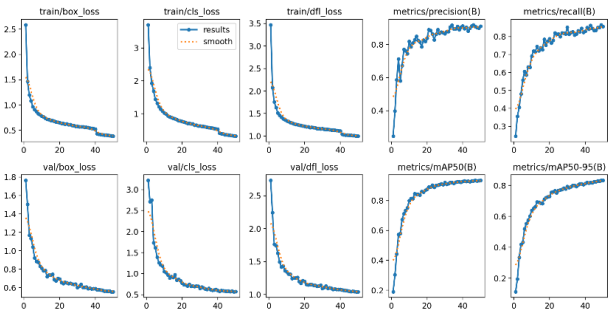

Fig.2- Training Metrics Plots of the Model

To ensure optimal detection performance, anchor box clustering was performed, analysing width, height, and aspect ratio distributions of objects in the dataset. The class-wise instance distribution is illustrated in **Fig. 3.**

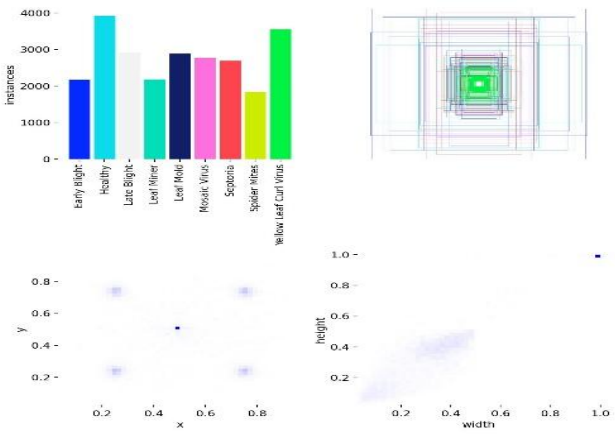

Fig.3 - Visualization of Class Frequencies and Spatial Distribution of Object Annotations

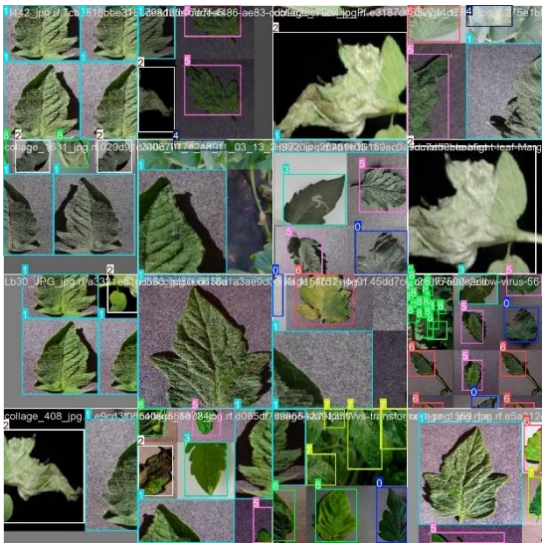

Fig. 4. - Anchor box clustering results

These visualizations guided the model configuration for improved bounding box predictions.

IV. RESULTS AND DISCUSSION

1. Model Performance Evaluation

The performance of the proposed YOLOv8-based ripeness detection model was evaluated using standard metrics including precision, recall, and F1-score. The evaluation was conducted on a test dataset comprising images of tomatoes captured under varied environmental conditions to simulate real-world scenarios.

1.A. Ripeness Detection Model

Table 1 – Ripeness Detection Model Performance

Class	Precision (%)	Recall (%)	F1-score (%)
Ripe	95.67	93.58	94.61
Unripe	93.89	88.84	91.29
Background	74.84	0.00	0.00

The confusion matrix for the ripeness detection model is shown in **Fig. 5**, which highlights the model's ability to correctly classify ripe and unripe tomatoes while identifying false positives and negatives.

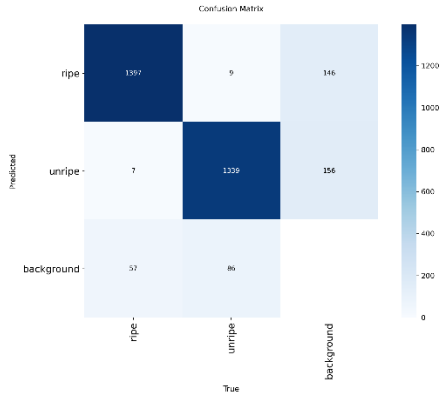

Fig.5 - The confusion matrix for the ripeness detection model

Sample inference output images demonstrating model detection of ripe and unripe tomatoes are presented in **Fig. 6**. Each bounding box is labelled with the predicted class and confidence score.

Fig. 6 – Sample Ripeness Detection Outputs.

1.B. Disease Detection Model:

The confusion matrix for the disease detection model is shown in **Fig. 7**, which highlights the model's ability to correctly classify different disease types and identify misclassifications.

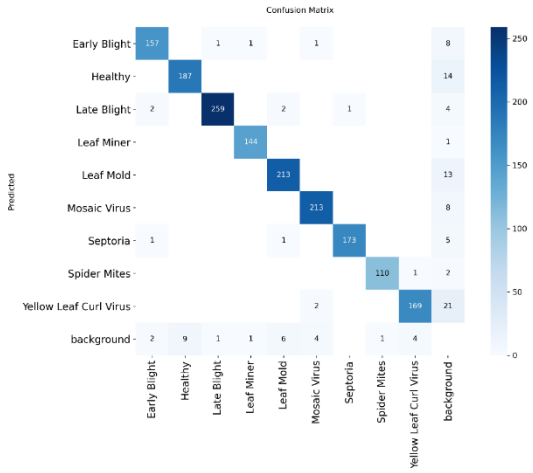

Fig.7 – Confusion Matrix for Disease Detection Mode

Disease Class	Precision (%)	Recall (%)	F1-score (%)
Early Blight	97.52	94.61	96.05
Healthy	94.92	93.03	93.96
Late Blight	99.23	97.37	98.29
Leaf Miner	97.96	99.31	98.63
Leaf Mold	96.85	94.21	95.51
Mosaic Virus	95.94	96.38	96.16
Septoria	98.86	96.63	97.73
Spider Mites	98.21	96.49	97.34
Yellow Leaf Curl Virus	96.59	88.48	92.36
Background	60.00	0.00	0.00

Table 2. –Disease Detection Model Performance

Detection of various tomato diseases are presented in **Fig. 8**. Each bounding box is labelled with the predicted disease class and confidence score.

Fig. 8 – Sample Disease Detection Outputs

B. Future Development

The current system demonstrates efficient ripeness and disease detection using YOLOv8 models with environmental monitoring. However, several enhancements can be implemented to further automate and optimize the agricultural process:

1. **Onboard Model Deployment:** By integrating an advanced embedded board such as the **NVIDIA Jetson Nano**, the detection models can run directly on the rover, eliminating the need for a separate laptop or external processing unit. This will significantly reduce system latency and improve real-time decision-making in the field.
2. **Robotic Arm Integration:** Attaching a **robotic manipulator** to the rover would allow automated harvesting of ripe tomatoes. The arm could identify, pick, and place the tomatoes with minimal human intervention, enhancing operational efficiency.
3. **Precision Agriculture Capabilities:** The rover can be equipped with targeted delivery systems for watering and pesticide application. This ensures that irrigation and pest control are applied **directly at the plant level**, reducing wastage and promoting sustainable farming practices.
4. **End-to-End Autonomous Farming:** Combining onboard detection, robotic harvesting, and precision treatment creates a **fully autonomous farming solution**, capable of monitoring crop health, harvesting ripe produce, and maintaining optimal growing conditions with minimal human oversight.

These future developments aim to transform the rover from a monitoring and detection system into a **fully functional, autonomous agricultural assistant**, capable of improving yield, reducing labour costs, and supporting sustainable farming practices.

V. CONCLUSION

This work presents an integrated agricultural monitoring and assistance system capable of detecting tomato ripeness and identifying multiple plant diseases using YOLOv8 models. The system achieved high performance across most target classes, with the ripeness detection model attaining a precision of 95.67% for ripe tomatoes and 93.89% for unripe tomatoes, while the disease detection model recorded precision values exceeding 94% for most classes. The deployment of these models, coupled with environmental sensor monitoring, enables timely and accurate decision-making in agricultural operations

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The proposed approach demonstrates the potential for improving crop quality assessment, optimizing resource utilization, and reducing manual labor in farming. Although the current system requires external computational support, it lays the groundwork for fully autonomous solutions. Future developments, such as deploying the models on embedded platforms like the NVIDIA Jetson Nano, integrating a robotic harvesting arm, and enabling precision irrigation and pesticide application, will further enhance the rover's capabilities, transforming it into a comprehensive, autonomous agricultural assistant.

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